Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Martin Byung-Guk Jun	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	2550 Northwestern Ave, west Lafayette, IN 47906			

2. Technology Overview

Title of Proposed Technology	**ERP-Integrated Autonomous Yard Operation Software for Trailer Auto Parking and Outbound Logistics**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Robotics/Automation Machinery		*Knowledge Services*	
	Medium Category	Small Category	Medium Category	Small Category
	Robot Control and Intelligence Technologies	*Robotics/Automation Machinery Software*	*Intelligent Enterprise Logistics Support Technologies*	*Distribution & Logistics Application Technologies*
Concept of Proposed Technology	**Core Concept & Potential Applications** - Develop software that connects ERP production and delivery orders with trailer movement inside factory and logistics yards. - Use ERP information such as trailer ID, production completion, and delivery orders to automatically generate trailer movement tasks. - Manage yard maps, parking areas, and parking availability to decide where trailers should be parked and when they should be retrieved. - Generate task-level commands for Autonomous Mobile Robots (AMRs) to pick up trailers, move them to parking spaces, or deliver them to outbound locations. - Keep track of the relationship between trailer IDs and their parking locations in real time.			

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	- After completing assigned tasks, guide AMRs to charging stations and keep them ready for the next instruction. - Operate on existing commercial AMR platforms without depending on specific robot hardware vendors. - Apply the software to trailer manufacturing plants, factory yards, and logistics yards where manual trailer handling is inefficient or unsafe.
Rationale for Joint R&D Project	- This project requires expertise in autonomous system control, large-scale logistics operation modeling, and optimization of robot task scheduling, which are areas where Purdue University is recognized as a leading research institution in the United States. **Why Global Collaboration?** Purdue has strong research capabilities in autonomous robotics, industrial engineering, and logistics systems, which directly match the technical challenges of yard operation orchestration, parking optimization, and multi-robot coordination. Futhermore, Purdue University, a leading overseas research institution, is a manufacturing-focused engineering university located in Indiana, a region surrounded by numerous manufacturing companies, and serves as an industrial hub in the United States. The university has already been collecting meaningful equipment data for smart manufacturing and digital transformation in the surrounding area and utilizing them for AI-based research. (1) Purdue maintains strong collaborative relationships with medium- to large-scale enterprises equipped with factories, facilities, and production lines; (2) it operates specialized research centers capable of conducting testbed-based research on manufacturing and physical AI; (3) it is strategically located in a major transportation hub in the U.S., making it the only academic institution where Korean companies can directly identify technological trends and market demands in the local environment. Therefore, global collaboration is essential. - The target application of this project is trailer manufacturing and yard operations in the United States, which requires technical validation and operational assumptions aligned with U.S. industrial environments. - This joint R&D structure allows the project to be conducted with technical rigor while remaining closely connected to practical deployment requirements in the U.S market.
Key Words	**1. Autonomous yard operation algorithms** **2. Trailer parking and retrieval control logic** **3. AMR task scheduling and dispatch algorithms** **4. Yard mapping and parking state modeling** **5. Multi-AMR coordination logic**

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	**R&D Objectives** The objective of this R&D project is to develop software technologies for autonomous yard operations that enable efficient trailer parking, retrieval, and outbound dispatch using Autonomous Mobile Robots (AMRs), and to establish a software-based orchestration framework applicable to large-scale factory and logistics yard environments. **Detailed Objectives** To achieve this objective, the project aims to advance and validate the developed software technologies through international joint R&D with a U.S.-based university, utilizing operational assumptions and yard scenarios relevant to the U.S. trailer and logistics market. Through simulation-based and pilot-scale validation, the project seeks to secure autonomous yard operation software technologies that can be applied to real industrial environments without reliance on proprietary AMR hardware. 1. Development of Core Software Technologies for Autonomous Yard Operations • Development of core software logic for autonomous trailer movement, including parking, retrieval, and outbound dispatch operations within factory and logistics yards • Design of yard-level decision-making logic that supports autonomous operation without manual intervention 2. Development of Yard Modeling and Parking Management Technologies • Modeling of yard layouts, parking zones, driving paths, and outbound points as structured software data models • Development of parking state management logic that maintains real-time mapping between trailer IDs and parking locations 3. Development of AMR Task Generation and Scheduling Technologies • Development of task generation logic that translates operational events into trailer movement tasks such as pick-up, parking, and delivery • Implementation of task scheduling and dispatch logic capable of coordinating multiple AMRs operating concurrently within the same yard 4. Development of Multi-AMR Coordination and Operation Control Logic • Development of coordination logic for managing conflicts, priorities, and task execution order among multiple AMRs			

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	• Implementation of operation control logic for post-task behaviors, including AMR return-to-charge and standby management 5. Development of Simulation-based Validation and Evaluation Technologies • Construction of simulation environments representing realistic factory and logistics yard conditions, including parking constraints and outbound demand scenarios • Evaluation of software performance and stability through scenario-based simulation and pilot-scale testing 6. Verification of Applicability to U.S. Trailer and Logistics Yard Environments • Validation of autonomous yard operation software under operational conditions representative of U.S.-based trailer manufacturing and logistics yards • Establishment of a technical foundation for future pilot deployment and commercialization in overseas markets
R&D Target Specifications	1. Autonomous Yard Operation Task Execution Reliability • Target Description: Develop autonomous yard operation software capable of executing trailer parking, retrieval, and outbound dispatch tasks through AMR coordination based on predefined yard operation logic. • Target Performance: Achieve a task execution success rate of 95% or higher across predefined simulation and pilot-scale scenarios, including sequential outbound orders and limited parking availability. 2. Yard Modeling and Parking State Management Accuracy • Target Description: Develop yard modeling and parking management software that represents yard layouts, parking zones, and trailer locations as structured data models and maintains real-time parking state information. • Target Performance: Maintain accurate mapping between trailer IDs and parking locations with state update latency of less than 1 second under simulated operational conditions. 3. Multi-AMR Task Scheduling and Coordination Performance • Target Description: Develop task scheduling and dispatch logic capable of coordinating multiple AMRs operating concurrently within the same yard environment, including conflict handling and priority control. • Target Performance: Support coordinated operation of at least three (3) AMRs without task collision or deadlock, verified through scenario-based simulation. 4. Trailer Movement Task Generation Latency • Target Description:

	Develop software logic that generates trailer movement tasks, including pick-up, parking, retrieval, and outbound delivery, based on operational events from production and logistics systems. • Target Performance: Generate and dispatch trailer movement tasks within 2 seconds after receiving operational events in simulation-based environments. 5. Integrated Software Validation and Technology Maturity Advancement • Target Description: Conduct integrated validation of autonomous yard operation software through simulation-based and pilot-scale demonstrations reflecting realistic factory and logistics yard conditions. • Target Performance: Achieve Technology Readiness Level (TRL) advancement from TRL 3 to TRL 6, and verify applicability through pilot-scale validation scenarios representative of U.S. trailer and logistics yard environments.
R&D Plan	This R&D project aims to secure the applicability of software technologies for autonomous yard operations that enable trailer parking, retrieval, and outbound dispatch using Autonomous Mobile Robots (AMRs). To this end, the project will be carried out in a phased manner, focusing on yard operation logic, task scheduling, multi-AMR coordination, and simulation-based validation. The R&D activities are structured into three phases: (1) core software technology development → (2) system integration and prototype development → (3) pilot-scale validation. **Phase 1:** Core Software Technology Design and Development (Achieving TRL 3–4) • Design of the overall software architecture for autonomous yard operations • Definition of yard data models, including yard layout, parking zones, driving paths, and outbound points • Design of trailer movement task models for parking, retrieval, and outbound dispatch • Design of baseline algorithms for task generation, parking allocation, and AMR scheduling **Phase 2:** System Integration and Software Prototype Development (Achieving TRL 5) • Implementation of core software modules for yard modeling, parking state management, and task execution control • Implementation of task scheduling and dispatch logic for multiple AMRs operating concurrently • Integration of software logic into a simulation environment representing factory and logistics yard conditions • Performance verification and refinement of software logic through scenario-based simulation

	Phase 3: Pilot-scale Validation and Technology Verification (Achieving TRL 6) • Execution of integrated simulation and pilot-scale validation under realistic yard operation scenarios • Verification of key performance indicators, including task execution success rate, scheduling performance, and system stability • Validation of applicability to U.S.-based trailer manufacturing and logistics yard environments • Preparation of technical documentation and validation results to support future deployment and commercialization

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	As the technical lead of this international joint R&D project, Purdue University serves as the overall technical coordinator, leading the research design, algorithm development, and validation framework for autonomous yard operation software based on Autonomous Mobile Robots (AMRs). Purdue University plans and manages the research components of the project with a focus on autonomous operation logic, task scheduling, multi-AMR coordination, and simulation-based validation, while supporting close technical collaboration with the industry partner to ensure applicability to real factory and logistics yard environments. Detailed Roles and Responsibilities 1. Overall Software Architecture Design and Technical Leadership • Design of the overall software architecture for autonomous yard operations, including yard modeling, task generation, and AMR coordination logic • Establishment of the research roadmap and definition of technical integration strategies for software modules 2. Development of Core Algorithms for Autonomous Yard Operations • Development of core algorithms for trailer movement operations, including parking, retrieval, and outbound dispatch • Design of decision-making logic for autonomous operation under dynamic yard conditions such as limited parking availability and sequential outbound demands 3. Research on Multi-AMR Task Scheduling and Coordination Logic • Research and development of task scheduling and dispatch algorithms for coordinating multiple AMRs operating concurrently within a shared yard • Development of conflict handling and priority control logic to ensure stable and efficient AMR operation

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	4. Design of Simulation-based Validation Frameworks • Design of simulation environments representing realistic factory and logistics yard layouts and operational scenarios • Definition of evaluation methodologies and performance metrics for validating autonomous yard operation software 5. Performance Verification and Technology Maturity Advancement • Validation of software performance through scenario-based simulation and pilot-scale testing • Verification of technology maturity advancement up to TRL 6 based on defined evaluation criteria • Identification of improvement areas and feedback to software design based on validation results 6. Coordination and Execution of International Joint R&D • Coordination of joint research activities and technical collaboration with the industry partner • Support for joint validation activities using operational scenarios relevant to U.S.-based trailer and logistics yards • Contribution to the dissemination of research outcomes and preparation of joint technical documentation
Expected R&R of Korean SME Partner	As a key implementation entity in the international joint R&D project, Miratek is responsible for the implementation, integration, and real-world validation of autonomous yard operation software developed through joint research. Based on the core algorithms and validation frameworks provided by the research institute, Miratek focuses on system integration, operational verification, and preparation for practical deployment in factory and logistics yard environments. Detailed Roles and Responsibilities 1. Software Implementation and System Integration for Autonomous Yard Operations • Implementation of autonomous yard operation software modules based on algorithms and technical specifications developed through joint research • Integration of yard modeling, parking management, task generation, and AMR coordination logic into a unified software system • Ensuring stable software operation under realistic factory and logistics yard conditions 2. Integration with Existing Systems and AMR Platforms • Integration of autonomous yard operation software with existing enterprise systems such as ERP and logistics management systems • Interface integration with existing commercial AMR platforms to enable task execution without modification of AMR hardware • Verification of operational compatibility between software logic and AMR control interfaces

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	3. Field-oriented Validation and Pilot Demonstration Support • Execution of simulation-based and pilot-scale validation in environments reflecting real factory and logistics yard operations • Verification of task execution reliability, scheduling performance, and system stability • Provision of operational feedback and improvement requirements to the research institute based on validation results 4. Review of Deployment Feasibility and Operational Constraints • Review of system deployment requirements considering yard layout diversity, operational constraints, and scalability • Analysis of software configuration, maintenance requirements, and operational costs for real-world application • Identification of constraints and risks related to on-site deployment and operation 5. Commercialization Preparation and Market Linkage Activities • Analysis of market requirements in trailer manufacturing and logistics yard automation sectors, particularly in the U.S. market • Definition of software-based service and business models derived from R&D outcomes • Collaboration with overseas partners to explore pilot deployment opportunities and PoC-based commercialization pathways

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1. Technological Outcomes** This project secures core software technologies for autonomous yard operations that enable trailer parking, retrieval, and outbound dispatch using Autonomous Mobile Robots (AMRs), and establishes a validated software framework applicable to large-scale factory and logistics yard environments. • Securing autonomous yard operation software technologies capable of achieving a task execution success rate of 95% or higher under realistic yard operation scenarios • Establishment of yard modeling, parking management, and task scheduling technologies that enable systematic and repeatable autonomous operations • Securing multi-AMR coordination and control logic validated through simulation-based and pilot-scale testing • Achieving system-level technological completeness of autonomous yard operation software at TRL 6 through integrated validation **2. Commercial and Economic Impact** The developed autonomous yard operation software represents a solution that can be directly

applied to trailer manufacturing plants and logistics yards, particularly in the United States, where many facilities operate with outdated manual processes and lack advanced automation solutions.

- Creation of new business opportunities for Korean companies in U.S. trailer manufacturing and logistics yard automation markets
- Establishment of software-based revenue models through system integration, deployment, and operation support services
- Expansion of overseas pilot demonstrations (PoC) in U.S. trailer factories preparing for facility modernization and automation investment
- Strengthening the competitiveness of Korean SMEs as early entrants in a market with limited existing solution providers

3. Industrial Impact

The project contributes to the advancement of intelligent manufacturing and logistics industries by introducing autonomous yard operation technologies into traditionally labor-intensive and manually operated environments.

- Improvement of operational efficiency and throughput in trailer manufacturing yards through automation of parking and outbound logistics processes
- Reduction of operational complexity and human dependency in large-scale yard management
- Acceleration of digital transformation in trailer manufacturing and logistics industries through software-centric automation approaches
- Establishment of a reference model for autonomous yard operation systems applicable to other industrial and logistics sectors

4. Strategic and Social Impact

By addressing structural inefficiencies in trailer yard operations, the project supports safer, more efficient, and more sustainable industrial logistics environments.

- Reduction of operational risks and safety incidents associated with manual trailer movement in factory yards
- Contribution to labor productivity improvement in aging and labor-constrained industrial environments
- Support for long-term sustainability of manufacturing and logistics operations through optimized asset utilization
- Strengthening international technology collaboration between Korean companies and U.S. research institutions, creating a foundation for continued joint innovation

6. References

(1) Boysen, Nils, et al. “Scheduling and routing of autonomous mobile robots in industrial environments: A survey.” *Resources, European Journal of Operational Research 287.2 (2020): 404–418.*

(2) McKinsey & Company. “Unlocking the industrial potential of robotics and automation.” *Resource, McKinsey & Company, 2023.*